Lecture 15

Nonlinear local evolution

A nonlinear equation need not have an explicit solution formula. We use the linear evolution to construct a nonlinear field by successive approximation. The construction gives a time interval shared by all data in a bounded ball.

Objectives. *We solve the one-dimensional ϕ^4 equation near a vacuum or a stationary kink. We prove the embedding and forcing estimates needed for contraction, then identify the only obstruction to continuing an energy solution.*

By the end of this lecture you will be able to:

- Bound a one-dimensional H^1 function in the supremum norm.
- Construct the nonlinear energy solution by a Duhamel contraction.
- Prove continuous dependence and the finite-lifespan blow-up alternative.

15.1 A stationary background and a perturbation

The potential and field equation are

$$V(\phi) = \frac{1}{2}(1-\phi^2)^2, \qquad \phi_{tt} - \phi_{xx} + V'(\phi) = 0, \qquad V'(u) = 2u(u^2-1).$$

Definition 15.1 (Stationary background). A *stationary background* is a time-independent solution b satisfying $-b'' + V'(b) = 0$. We use

$$b(x) \in \{1, -1, \tanh x\}.$$

The constant minima 1 and -1 are the *vacua*. The profile $b(x) = \tanh x$, connecting these vacua, is the *kink*.

For a vacuum the stationary equation is immediate. For the kink, $b' = 1 - b^2$ and $b'' = -2b(1-b^2) = V'(b)$. All three backgrounds obey $|b| \le 1$. Figure 15.1 shows their profiles. None belongs to $L^2(\mathbb{R})$.

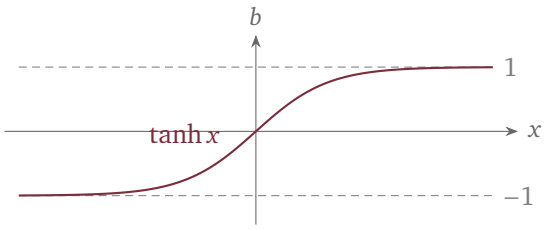


Figure 15.1: The two constant vacua and the kink $b(x) = \tanh x$. The kink approaches -1 and 1 at the two spatial ends.

Set $\phi = b + v$. Subtract the stationary equation before placing any term in L^2. Using a linear Klein–Gordon operator of mass one gives

$$\begin{aligned} v_{tt} - v_{xx} + v &= N_b(v), \\ N_b(v) &= v - [V'(b+v) - V'(b)] = (3 - 6b^2)v - 6bv^2 - 2v^3. \end{aligned} \tag{15.1}$$

There is no background forcing term. The reference mass one is a choice for the integral equation. The physical linearisation at either vacuum has $V''(\pm 1) = 4$, hence mass two.

Use the energy space $\mathcal{E} = H^1(\mathbb{R}) \times L^2(\mathbb{R})$ of Lecture 4, with the equivalent Hilbert norm

$$\|(f,g)\|_{\mathrm{lin}}^2 = \|f\|_{H^1}^2 + \|g\|_2^2.$$

An *energy solution* has $v \in C([0,T]; H^1) \cap C^1([0,T]; L^2)$, the prescribed data, and satisfies (15.1) distributionally on $(0,T) \times \mathbb{R}$.

15.2 The one-dimensional estimate

Lemma 15.2 (Embedding and nonlinear bound). *Each $v \in H^1(\mathbb{R})$ has a continuous bounded representative, with*

$$\|v\|_\infty \le \frac{1}{\sqrt{2}} \|v\|_{H^1}. \tag{15.2}$$

For $\|v\|_{H^1}, \|w\|_{H^1} \le R$, the force is in L^2 and obeys

$$\|N_b(v) - N_b(w)\|_2 \le L_R \|v - w\|_{H^1}, \qquad L_R = 3 + 6\sqrt{2}R + 3R^2. \tag{15.3}$$

In particular, $N_b : H^1 \to L^2$ is locally Lipschitz and $N_b(0) = 0$.

Proof. The unitary inverse Fourier formula and Cauchy–Schwarz give

$$|v(x)| \le (2\pi)^{-1/2} \int |\widehat{v}(k)| \, \mathrm{d}k \le (2\pi)^{-1/2} \left(\int \frac{\mathrm{d}k}{1+k^2} \right)^{1/2} \|v\|_{H^1} = \frac{1}{\sqrt{2}} \|v\|_{H^1}.$$

Thus $\widehat{v} \in L^1$. Its inverse is continuous by dominated convergence and represents the same L^2 function.

Put $M = R/\sqrt{2}$. Both $|v|$ and $|w|$ are at most M. Factor the differences of their squares and cubes in (15.1). Since $|3 - 6b^2| \le 3$, the pointwise bound is

$$|N_b(v) - N_b(w)| \le (3 + 12M + 6M^2)|v - w|.$$

Taking L^2 norms proves (15.3). Setting $w = 0$ also proves that every term in $N_b(v)$ is in L^2. ∎

15.3 The forced linear formula

Let $U(t) : \mathcal{E} \to \mathcal{E}$ be Lecture 4's mass-one evolution, including its velocity. Its Fourier frequency is $\nu(k) = \sqrt{1 + k^2}$. In Fourier variables the pair $(\nu \widehat{f}, \widehat{g})$ rotates by angle $t\nu$. Consequently

$$\|U(t)z\|_{\mathrm{lin}} = \|z\|_{\mathrm{lin}}, \qquad U(t+s) = U(t)U(s).$$

The rotation identity and the group law follow from the sine and cosine addition formulas. Strong continuity is the one proved in Lecture 4.

Definition 15.3 (Duhamel formula and mild solution)**.** For $F \in C([0,T];L^2)$, its zero-data *Duhamel pair* is

$$(\mathcal{D}F)(t) = \int_0^t U(t-s)(0,F(s))\,\mathrm{d}s. \tag{15.4}$$

A *mild solution* with data $z_0 = (v_0, g_0) \in \mathcal{E}$ is a continuous pair $z = (v,p) : [0,T] \to \mathcal{E}$ satisfying

$$z(t) = U(t)z_0 + \mathcal{D}(N_b(v))(t). \tag{15.5}$$

The integral is a norm limit of Riemann sums in the complete energy space. Continuity of its integrand makes these sums Cauchy by uniform continuity. Its first component is the Fourier sine formula

$$\widehat{(\mathcal{D}F)_1}(t,k) = \int_0^t \frac{\sin((t-s)\nu(k))}{\nu(k)}\widehat{F}(s,k)\,\mathrm{d}s,$$

understood as an L^2 identity. The sign is positive because the right-hand side of (15.1) is $+N_b(v)$.

Lemma 15.4 (Forced energy estimate)**.** *For continuous L^2 forcing, $\mathcal{D}F = (w,p)$ is continuous into $\mathcal{E}$, $w_t = p$ in L^2, and $w_{tt} - w_{xx} + w = F$ weakly, with zero initial data. Moreover,*

$$\|\mathcal{D}F(t)\|_{\mathrm{lin}} \le \int_0^t \|F(s)\|_2\,\mathrm{d}s. \tag{15.6}$$

Every energy solution of this forced equation is its homogeneous solution plus this Duhamel pair.

Proof. Isometry of U and the triangle inequality for Riemann sums prove the bound. Strong continuity of U gives continuity of the integral.

To justify the equation, use the smooth compact Fourier cutoffs P_j of Lecture 4 and set $F_j = P_jF$. These converge uniformly in $C([0,T];L^2)$: the compact set $F([0,T])$ has finite nets, and bounded operators converging strongly converge uniformly on each such compact set. For bounded frequencies, $1 - \partial_x^2$ is a bounded L^2 operator. Differentiating the sine and cosine integrals gives $(w_j)_t = p_j$ and $(p_j)_t = \partial_x^2 w_j - w_j + F_j$. Thus the cutoff pair satisfies the forced equation.

The bound applied to $F_j - F$ gives convergence of the pairs uniformly in $\mathcal{E}$. Passing to the limit in $w_j(t) = \int_0^t p_j(s)\,\mathrm{d}s$ proves $w_t = p$ in L^2. Against every compact smooth test ψ in $(0,T)\times\mathbb{R}$, the cutoff equation is

$$\int w_j(\psi_{tt} - \psi_{xx} + \psi)\,\mathrm{d}t\,\mathrm{d}x = \int F_j\psi\,\mathrm{d}t\,\mathrm{d}x.$$

Uniform L^2 convergence passes both integrals to the limit. The initial data also converge and are zero.

Finally, subtract this constructed pair and the homogeneous solution from any energy solution with the same forcing and data. The difference has zero data and solves the homogeneous equation. Lecture 4's uniqueness proves it is zero. Its frequency-cutoff proof applies on $[0,T]$ as well: it integrates the cutoff energy from the continuous initial traces at zero. ∎

Lemma 15.2 makes $N_b(v)$ continuous into L^2 for every energy solution. Lemma 15.4 therefore proves that mild and energy solutions of the nonlinear equation are the same, with $p = v_t$.

15.4 Contraction and continuation

The *lifespan* T_* is the supremum of the times to which the unique solution with the given data can be continued in the energy class.

Theorem 15.5 (Local nonlinear evolution). *For each allowed b and every $z_0 = (v_0, g_0) \in \mathcal{E}$, there is a unique energy solution on some interval $[0, T]$. A common choice for all $\|z_0\|_{\mathrm{lin}} \le B$, with $B \ge 1$, is*

$$R = 2B, \qquad T = \frac{1}{2L_R}. \tag{15.7}$$

On this interval the solution norm is at most R, and the solution map is Lipschitz in $C([0,T];\mathcal{E})$. The maximal solution satisfies

$$T_* < \infty \quad \Longrightarrow \quad \|(v(t), v_t(t))\|_{\mathrm{lin}} \longrightarrow \infty \quad \text{as } t \uparrow T_*. \tag{15.8}$$

Time reversal gives the same local and continuation statements for negative times.

Proof. The path space $X_T = C([0,T];\mathcal{E})$ is complete in the supremum of the linear norm, by completeness of $\mathcal{E}$. Its closed ball $\mathcal{B}_R$ is nonempty. Define Φ on this ball by the right-hand side of (15.5). The preceding lemmas give

$$\|\Phi z\|_{X_T} \le B + TL_R R = R, \qquad \|\Phi z - \Phi \widetilde{z}\|_{X_T} \le TL_R \|z - \widetilde{z}\|_{X_T} = \frac{1}{2}\|z - \widetilde{z}\|_{X_T}.$$

The Banach contraction principle from *Analysis* I, Lecture 1, says that a contraction of a nonempty closed subset of a Banach space has a unique fixed point. It supplies the mild solution here, and hence the energy solution.

For two data in the same ball, subtract their integral equations and take the supremum norm. This gives

$$\|z - \widetilde{z}\|_{X_T} \le \|z_0 - \widetilde{z}_0\|_{\mathrm{lin}} + \frac{1}{2}\|z - \widetilde{z}\|_{X_T},$$

so the difference is at most twice the initial difference. Every other energy solution stays inside the construction ball for a sufficiently short initial interval, by continuity. It agrees there with the fixed point. The same argument can be restarted at any common time. The set of agreement is closed by continuity and open by this local uniqueness, so the solutions agree throughout their common interval. Repeated short intervals also give continuous dependence on every compact interval on which the solutions remain in a common bounded ball.

Unique local solutions glue to a maximal one. Suppose $T_* < \infty$ and there were times $t_j \uparrow T_*$ with solution norm at most B. Replace B by $\max\{B, 1\}$. Restarting at t_j gives a solution on $[t_j, t_j + 1/(2L_{2B})]$, with the same positive duration for every j. For large j this extends beyond T_* and agrees on the overlap by uniqueness, a contradiction. Thus every bounded ball is eventually left, which proves (15.8). ∎

Example 15.6 (A common contraction time near a vacuum). Take $b = 1$ and data with $\|(v_0, g_0)\|_{\mathrm{lin}} \le 1$. The force is $N_1(v) = -3v - 6v^2 - 2v^3$. The construction uses

$$R = 2, \qquad L_2 = 15 + 12\sqrt{2}, \qquad T = \frac{1}{30 + 24\sqrt{2}}.$$

Its radius bound is $1 + TL_2 \cdot 2 = 2$, and its contraction factor is $TL_2 = 1/2$. These constants prove a common positive time for the entire data ball. They do not estimate the maximal lifespan sharply.

15.5 Why the dimension matters

The embedding used above fails in higher dimensions. In three dimensions, let $\eta(x) = e^{-|x|^2}$ and $u_\lambda(x) = \lambda^{1/2}\eta(\lambda x)$ for $\lambda \geq 1$. With $I = (\pi/2)^{3/2}$, substitution gives

$$\|u_\lambda\|_{H^1}^2 = I(\lambda^{-2} + 3) \leq 4I, \qquad \|u_\lambda\|_\infty = \lambda^{1/2} \longrightarrow \infty.$$

Thus an H^1 bound cannot replace the supremum bound in that proof. This does not disprove cubic local existence in three dimensions: $H^1(\mathbb{R}^3) \subset L^6(\mathbb{R}^3)$ controls a cubic in L^2, as developed on Sheet 15. Higher dimensions require estimates suited to the nonlinearity.

In two dimensions the failure is logarithmic. For $\Lambda \geq e$, the Fourier profile

$$\widehat{u}_\Lambda(k) = \frac{\mathbf{1}_{\{1 \leq |k| \leq \Lambda\}}}{\sqrt{\log \Lambda}\,|k|^2}$$

gives, by polar integration and the unitary inverse transform,

$$\|u_\Lambda\|_{H^1}^2 = 2\pi\left[1 + \frac{1 - \Lambda^{-2}}{2\log\Lambda}\right] \leq 3\pi, \qquad u_\Lambda(0) = \sqrt{\log \Lambda}.$$

The norm calculation integrates $(1 + r^2)/r^3$ over $1 \leq r \leq \Lambda$. The value at zero uses $\int_1^\Lambda dr/r = \log\Lambda$. The one-dimensional proof therefore cannot be applied unchanged even in two dimensions. Lecture 16 stays in one dimension and uses energy to control the continuation norm.

Summary. Subtracting a stationary vacuum or kink gives an $H^1 \times L^2$ perturbation equation without a background force. In one dimension the Fourier embedding makes its cubic force locally Lipschitz from H^1 to L^2. The forced Klein–Gordon estimate gives a contraction on energy paths, uniformly on bounded data balls. Mild and weak energy solutions coincide. Uniqueness permits continuation, and any finite maximal lifespan forces the perturbation norm to tend to infinity.

Self-check. Why is $V'(b + v)$ not used alone as the L^2 force around the kink? What happens when $v = 0$ in the correct perturbation equation?

Answer. The stationary term must cancel b''. The correct difference is $V'(b + v) - V'(b)$, so $N_b(0) = 0$ and the zero perturbation remains zero. The construction needs no assumption that $b \in L^2$.

Self-check. For the unit data ball in the worked example, what happens to the proved contraction factor if the chosen time is doubled?

Answer. It becomes $2TL_2 = 1$. The stated estimate no longer gives a strict contraction, even though a longer actual solution may exist.

Self-check. Does a finite maximal lifespan give only an unbounded sequence of energy-space norms, or does the norm tend to infinity?

Answer. The energy-space norm tends to infinity as $t \uparrow T_*$. A bounded subsequence would supply a uniform restart duration and extend the solution beyond T_*.

Self-check. Does the three-dimensional concentration example prove that the cubic wave equation has no $H^1 \times L^2$ local theory?

Answer. No. It disproves the $H^1 \to L^\infty$ estimate used here. The separate embedding $H^1 \to L^6$ can control the cubic in L^2.